

MC10182/MC10582

2-BIT ARITHMETIC LOGIC UNIT and FUNCTION GENERATOR

Function Select		POSITIVE LOGIC	
		Logic Function M is High F	Arithmetic Operation M is Low F
L	L	$F = A \odot B$	$F = A \text{ plus } B \text{ plus Carry}$
L	H	$F = A \oplus B$	$F = \bar{A} \text{ plus } B \text{ plus Carry}$
H	L	$F = A \bullet B$	$F = A \text{ plus } \bar{B} \text{ plus Carry}$
H	H	$F = A + B$	$F = A \text{ times } 2$

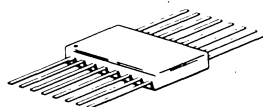
$P_D = 575 \text{ mW typ/pkg (No Load)}$
 $t_{pd} = 7.5 \text{ ns typ (A or B to F or } C_{n+2})$
 $= 2.7 \text{ ns typ (} C_n \text{ to } C_{n+2} \text{ or F)}$
 $= 6.5 \text{ ns typ (A to } P_G \text{ or } G_G)$



P SUFFIX
PLASTIC PACKAGE
CASE 648
MC10182 only



L SUFFIX
CERAMIC PACKAGE
CASE 620



F SUFFIX
CERAMIC PACKAGE
CASE 650
MC10582 only

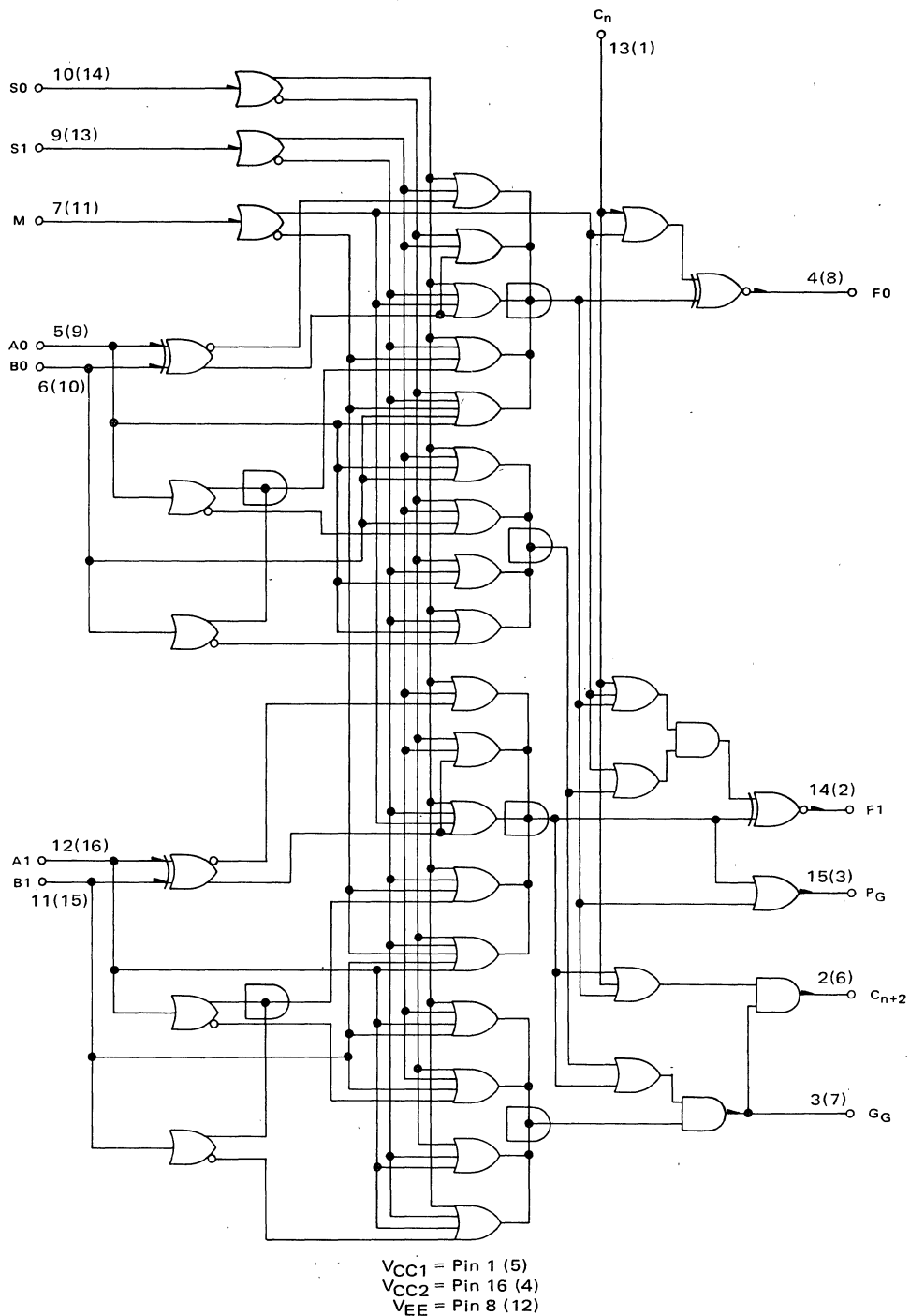
The MC10182/MC10582 is a high-speed arithmetic logic unit capable of performing 4 logic operations and 4 arithmetic operations on two 2-bit words. Full internal carry is incorporated for arithmetic operation.

Arithmetic logic operations are selected by applying the appropriate binary word to the select inputs (S0 and S1) as indicated in the tables of arithmetic/logic functions. Group carry propagate (P_G) and carry generate (G_G) are provided for a second order look ahead carry using the MC10179. The internal carry is enabled by applying a low level voltage to the mode control input (M).

The MC10182 provides an alternate to the MC10181 four-bit ALU for applications not requiring the extended functions of the MC10181 or for applications requiring a 16-pin package. The MC10182 also differs from the MC10181 in that Word A and Word B are treated equally for addition and subtraction (A plus B, A minus B, B minus A).

Characteristic	Symbol	-55°C		-30°C		+25°C		+85°C		+125°C		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Power Supply Drain Current	I_E	—	152	—	152	—	138	—	152	—	152	mAdc
Input Current	I_{inH}	—	375	—	350	—	220	—	220	—	220	μAdc
Pins 7,9,10		—	660	—	620	—	390	—	390	—	390	
Pins 5,12		—	495	—	460	—	290	—	290	—	290	
Pins 6,11		—	595	—	560	—	350	—	350	—	350	
Pin 13		—	595	—	560	—	350	—	350	—	350	
Switching Times												ns
Propagation Delay	t_{pd}											
C_n to C_{n+2} or F		1.5	6.1	1.5	5.9	1.5	5.6	1.6	6.2	1.6	6.6	
M or S to F; A or B to P_G or G_G		2.3	10.8	2.3	10.5	2.3	10	2.4	11	2.4	11.7	
A0 or B0 to F; A1 or B1 to F1		2.3	10.8	2.3	10.5	2.3	10	2.4	11	2.4	11.7	
A0, B0, or A1 to C_{n+2}		2.3	10.8	2.3	10.5	2.3	10	2.4	11	2.4	11.7	
B1 to C_{n+2}		2.8	13	2.8	12.6	2.8	12	2.9	13.2	2.9	14	
Rise Time, Fall Time (20% to 80%)	t_r, t_f	1.5	4.9	1.5	4.7	1.5	4.5	1.6	5.0	1.6	5.3	ns

-55°C and +125°C test values apply to MC105xx devices only.



Numbers at ends of terminals denote pin numbers for L and P packages.
 Numbers in parenthesis denote pin numbers for F package.

These outputs are not normally used during logic operation.